

Development, simulation & implementation of precoding techniques for satellite links

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Ghent, May 2026

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Acknowledgements

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by

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Faculty of Engineering and Architecture

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Abstract — Laboris magna irure quis sit et veniam voluptate duis elit esse do excepteur. Officia laboris adipisicing laboris duis ut cillum aliquip. Do veniam aute non do cillum sit pariatur consectetur adipisicing qui. Lorem Lorem veniam minim exercitation cupidatat tempor do aute dolore. Nostrud adipisicing tempor quis qui fugiat sint quis ut aute eiusmod Lorem enim veniam. Elit labore Lorem do proident amet anim exercitation culpa enim cillum qui. Aliquip reprehenderit commodo labore minim laborum mollit ut veniam in irure. Minim sit in non aliqua ut elit veniam irure pariatur non velit consequat. Cupidatat veniam occaecat culpa eu irure aute consectetur dolore culpa deserunt quis officia do. Aliqua aliqua eu velit excepteur culpa ea dolore occaecat. Ea est cillum nulla ea cupidatat esse adipisicing ut. Cillum amet esse ex ex enim elit excepteur ullamco in qui aliqua Lorem veniam. Sunt consectetur aliquip commodo exercitation pariatur id tempor adipisicing aliqua. Fugiat cillum ea irure laborum consectetur ipsum. Excepteur laboris eu veniam ex ad consequat esse id ullamco ea cupidatat ad nisi. Aute nisi velit excepteur adipisicing dolore pariatur sunt. Dolore mollit cupidatat esse sint ipsum est nulla nisi eu cupidatat excepteur sint cillum. Culpa irure magna culpa eiusmod ut nisi ipsum veniam elit est occaecat veniam. Sunt adipisicing eu minim qui ea laboris ullamco ut. Aliquip occaecat in occaecat ut tempor dolore sunt consequat qui cillum reprehenderit. Voluptate commodo cupidatat ullamco sint aliquip veniam consequat do. Nisi velit excepteur aute laboris aliqua fugiat minim occaecat quis non incididunt.

Keywords — Precoding Techniques, High-Troughput Satellite Communications, MIMO, Outdated CSI

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Fig. 1. Example of a figure caption.

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ACKNOWLEDGMENT

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- [1] A. I. Perez-Neira, M. A. Vazquez, M. B. Shankar, S. Maleki, and S. Chatzinotas, “Signal processing for high-throughput satellites: Challenges in new interference-limited scenarios,” *IEEE Signal Processing Magazine*, vol. 36, no. 4, pp. 112–131, 2019.

Contents

Acknowledgements	i
Abstract	ii
Extended Abstract	iii
Contents	vi
List of Figures	viii
Symbols and Notations	ix
Abbreviations	ix
System Parameters	xi
Symbols	xi
 I Background Material	 1
 II Precoding for Satellite Communication Systems	 2
 1 High-Throughput Satellite Communication Systems	 3
1.1 Motivation	3
1.2 Modeling Aspects	4
1.3 Classification	6
 2 Precoding under Outdated CSI	 9
2.1 Problem Formulation	9
2.2 Impact of Outdated CSI	10
2.3 Compensation for Outdated CSI	12
2.3.1 Compensation Strategy	13
2.3.2 Parameter Optimization	14
2.3.3 Performance Evaluation	15
2.4 Evaluation on a More Realistic Channel	17
2.4.1 Channel Model	17
2.4.2 Performance Evaluation	19

<i>CONTENTS</i>	vii
2.5 Conclusion	20
2.6 Conclusion	21
III Appendices	23

List of Figures

1.1	Multibeam HTS communication system architecture	4
2.1	Impact of outdated CSI on the achievable sum-rate (ZF and BD precoding) . . .	11
2.2	Impact of outdated CSI on the achievable sum-rate (WMMSE precoding)	12
2.3	Impact of the pilot rate on the achievable sum-rate with AR-based compensation	14
2.4	Impact of the context window length on the achievable sum-rate with AR-based compensation	15
2.5	Impact of the proposed compensation technique on achievable sum-rate (ZF and BD)	16
2.6	Comparison of perfect, outdated and compensated CSI for WMMSE precoding .	17
2.7	Uniform planar array	18
2.8	Impact of Outdated and Predicted CSI on the BER (ZF precoding)	20

Symbols and Notations

Abbreviations

acf	autocorrelation function
AoA	angle of arrival
AoD	angle of departure
AR	autoregressive
AWGN	additive white Gaussian noise
BS	base station
BD	block diagonalization
BER	bit error rate
CS	circularly symmetric
CSI	channel state information
DFT	discrete Fourier transform
DPC	dirty paper coding
EVD	eigenvalue decomposition
FFT	fast Fourier transform
FIR	finite impulse response
GEO	geostationary earth orbit
GW	gateway
HTS	high-throughput satellite
IBR	information bit rate
i.i.d.	independent and identically distributed
KKT	Karush-Kuhn-Tucker
LEO	low earth orbit

LoS	line-of-sight
LSV	left singular vector
MI	mutual information
MIMO	multiple-input multiple-output
MMSE	minimum mean squared error
MSE	mean squared error
MSS	mobile satellite services
MU	multi-user
NLoS	non-line-of-sight
pdf	probability density function
PAM	pulse amplitude modulation
PSD	power spectral density
PSK	phase shift keying
QAM	quadrature amplitude modulation
QoS	Quality-of-Service
RTT	round-trip time
RX	receiver
SINR	signal-to-interference-plus-noise ratio
SISO	single-input single-output
SNR	signal-to-noise ratio
SR	sum rate
SU	single-user
SVD	singular value decomposition
THP	Tomlinson–Harashima precoding
TX	transmitter
UPA	uniform planar array
UT	user terminal
WMSE	weighted mean squared error
WMMSE	weighted minimum mean squared error
WSR	weighted sum rate

ZF zero-forcing

System Parameters

K	Number of UTs
N_T	Number of transmit antennas at the BS
N_R	Number of receive antennas at each UT
N_S	Number of data streams per UT
P_t	Total transmit power
N_0	Noise power on each receive antenna

Symbols

SU-MIMO

MU-MIMO TO DO

Part I

Background Material

Part II

Precoding for Satellite Communication Systems

Chapter 1

High-Throughput Satellite Communication Systems

The preceding chapters developed the theory of linear precoding for generic single-user (SU)- and multi-user (MU)-multiple-input multiple-output (MIMO) systems. Before turning to the specific problem addressed by this work, the present chapter situates these results within the context of satellite communications. We first motivate the use of precoding in modern high-throughput satellite (HTS) systems and describe the system architecture considered in the remainder of this work. We then discuss the channel features and modeling assumptions that distinguish the satellite setting from the terrestrial MU-MIMO scenario of Chapter ???. Finally, we review the main families of precoding techniques proposed in the literature and identify the specific category that is relevant for our work.

1.1 Motivation

Satellite communications have recently regained considerable attention as a key enabler of ubiquitous broadband connectivity. They provide cost-efficient coverage of large geographical areas, including regions where deploying a terrestrial infrastructure is impractical, and are envisioned as an integral component of future 5G and beyond networks [1, 2].

To meet the rapidly growing demand for data rates, modern satellite systems have evolved from single-beam architectures toward high-throughput satellite (HTS) systems, in which the coverage area is partitioned into many narrow spot beams. This multi-beam architecture enables spatial multiplexing by concurrently delivering independent data streams to geographically separated users. In conventional designs, inter-beam interference is managed through partial frequency reuse and polarization isolation, whereby adjacent beams are assigned non-overlapping frequency sub-bands or orthogonal polarizations. While effective at reducing interference, this approach significantly constrains spectral efficiency, since the full available bandwidth cannot be reused across all beams simultaneously. On the other hand, full frequency reuse maximizes the spectrum available per beam. The price to pay is a substantial increase in inter-beam interference: signals intended for one beam leak into adjacent beams through the sidelobes of the antenna radiation

pattern, and the system transitions from a noise-limited regime to an interference-limited one. Precoding has emerged as the natural solution: by exploiting channel state information (CSI) at the transmitter, interference can be suppressed while the available spectrum is reused across all beams [1].

1.2 Modeling Aspects

System Architecture

An illustration of the system architecture of a multi-beam satellite system is shown in Figure 1.1. It consists of three main entities: the gateway (GW), the satellite or base station (BS), and the user terminals (UTs). The GW is connected to the core network and usually performs all baseband processing. The link between the GW and the satellite is referred to as the feeder link, while the link between the satellite and the UTs is referred to as the user link. The direction from the GW through the BS to the UTs is called the forward link, while the opposite direction is called the return link. The forward link is by far the most rate-demanding direction, and centralizing the precoder design at the GW (or BS) aligns naturally with the MU-MIMO downlink framework developed in Chapter ??.

Throughout this work, we focus on the forward link of a multi-beam HTS system serving mobile UTs.

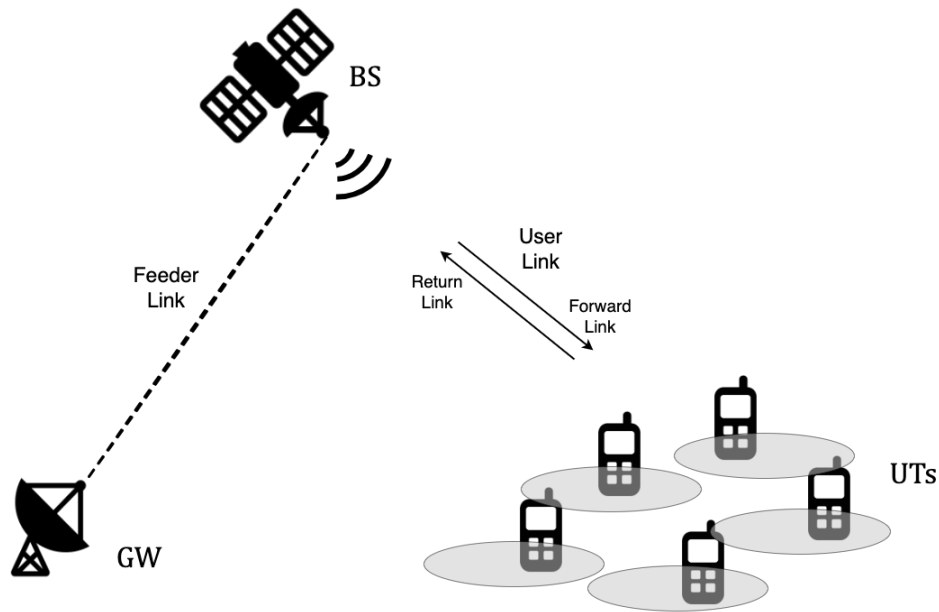


Figure 1.1: Schematic view of the multibeam HTS communication system architecture.

Features of Satellite Communications

Compared to terrestrial MU-MIMO systems, satellite communications exhibit several distinctive features. The propagation conditions are dominated by a line-of-sight (LoS) component between the satellite and the UTs, accompanied by a weaker scattered contribution arising from the local environment of the UT [3]. The free-space path loss is several orders of magnitude larger than

in terrestrial links, and the received power can be further reduced by atmospheric impairments such as rain attenuation. For mobile satellite services, however, communication often takes place in the L- or S-band, where propagation losses are smaller than in the Ka-band commonly used for fixed satellite services [1]. This comes at the expense of a more limited available bandwidth, which further motivates aggressive frequency reuse. For the present work, the two most relevant features are therefore the mobility of the UTs and the large propagation delay.

The mobility of the UTs and/or the satellite makes the channel time-varying through Doppler effects. For geostationary earth orbit (GEO) satellites, which appear essentially stationary relative to the Earth's surface, the satellite-induced Doppler shift is negligible, and the dominant source of channel variation is UT motion alone. For low earth orbit (LEO) satellites, the high orbital velocity produces a substantial satellite-induced Doppler shift. However, since the satellite's trajectory is known with high accuracy, this contribution is deterministic and can be pre-compensated (see e.g. [4]). The residual, more difficult component to track is the Doppler shift induced by the motion of the UT, since it depends on the instantaneous user velocity and propagation geometry. So, in both cases, UT mobility is the primary driver of channel time-variation, and it causes the channel to fluctuate at a rate governed by the UT speed and the carrier frequency.

The large geographical distances between the GW, the satellite, and the UTs introduce a propagation delay that is several orders of magnitude larger than in terrestrial systems. For a GEO satellite, which orbits at an altitude of approximately 35,786 km, the one-way propagation delay is on the order of 250 ms. Even for a LEO constellation, the round-trip time (RTT) remains in the tens of milliseconds. Since the precoder is computed based on CSI provided by the UTs, this delay implies that the channel used to design the precoder is inevitably outdated relative to the channel experienced when the precoded signal reaches the UTs. In a static scenario, this mismatch is of limited importance. In combination with a time-varying channel, however, it forms a major challenge for precoding design.

Channel Model

No single probabilistic model captures the full range of propagation conditions encountered in mobile satellite communications, and a variety of models have been proposed in the literature for different scenarios.

In this work, we begin with a deliberately simplified model in which the channel of UT_k is represented as a frequency-flat Rician fading channel [3, 5]:

$$\mathbf{H}_k(t) = \sqrt{\frac{\kappa}{\kappa + 1}} \mathbf{H}_k^{\text{LoS}}(t) + \sqrt{\frac{1}{\kappa + 1}} \mathbf{H}_k^{\text{NLoS}}(t), \quad (1.1)$$

where $\kappa \geq 0$ is the Rician factor, $\mathbf{H}_k^{\text{LoS}}(t)$ is the deterministic line-of-sight (LoS) component, and $\mathbf{H}_k^{\text{NLoS}}(t)$ is the time-varying stochastic non-line-of-sight (NLoS) component.

The Rician factor quantifies the power ratio between the LoS and the NLoS contribution. It typically takes values around 5 dB in urban environments and up to 15 dB or more in rural environments. In this work, it is set to 5 dB.

The LoS component is treated as time-invariant, and the small spatial variations across the antenna arrays are neglected. So, all entries of $\mathbf{H}_k^{\text{LoS}}$ share a common, constant phase offset θ_k associated with UT k :

$$\mathbf{H}_k^{\text{LoS}} = e^{j\theta_k} \mathbf{1}_{N_R \times N_T} \quad (1.2)$$

As a consequence, the LoS component does not contain any temporal or spatial variation.

The NLoS component $\mathbf{H}_k^{\text{NLoS}}(t)$ captures the scattered multipath contribution. Each entry is considered independent and identically distributed (i.i.d.), and modeled as a zero-mean unit-variance complex Gaussian random variable consistent with the Rayleigh fading model as described in Appendix ???. The temporal correlation of the NLoS entries is governed by the Doppler spread induced by UT mobility. Under the classical isotropic scattering assumption, the power spectral density of each channel entry follows the Jakes model:

$$S_{\mathbf{H}_k^{\text{NLoS}}}(f) = \frac{1}{\pi f_D \sqrt{1 - (f/f_D)^2}}, \quad |f| \leq f_D, \quad (1.3)$$

where $f_D = vf_c/c$ is the maximum Doppler frequency, determined by the UT speed v , the carrier frequency f_c , and the speed of light c . Further details on the Doppler power spectrum are provided in Appendix ??, and the generation of a time-correlated random process with a prescribed autocorrelation function or power spectral density is addressed in Appendix ??.

The model as stated rests on two simplifying assumptions: spatial correlation between channel entries is neglected, and the LoS Doppler shift is assumed to be perfectly pre-compensated. While these assumptions yield a tractable and well-understood baseline, they do not always hold in practice. At the end of the following chapter, we therefore describe how this model can be extended to a more realistic setting that accounts for spatial correlation and a residual LoS Doppler component, and we briefly evaluate the impact of these features on the performance of the proposed precoding techniques.

1.3 Classification

A large number of precoding techniques have been proposed for satellite communications, differing in the assumptions they make about the system and in the design objectives they pursue. A comprehensive survey can be found in [6]. The objective of this section is not to reproduce this overview, but rather to position the present work within the broader landscape and to identify the techniques that are most closely related.

A Taxonomy of Precoding Techniques

Besides the distinction between linear and nonlinear techniques, existing satellite precoding methods can be classified along several other non-exclusive axes. For each axis we briefly recall the main alternatives and indicate the choice made in this work.

A first distinction concerns the traffic model. In practical multibeam satellite systems, a single beam may simultaneously serve multiple UTs within the same frame, which leads naturally to a

multi-group multicast formulation. To avoid additional complications introduced by multicast scheduling, we restrict the analysis to the unicast case, in which only a single UT is served per beam. A second axis concerns the optimization criterion. Depending on the system objective, the precoder may be designed to maximize the sum rate under a transmit power constraint, or to minimize the transmit power provided that Quality-of-Service (QoS) metric is met. Closely related to this is the choice of power constraint. In practical satellite payloads, each feed is typically driven by its own power amplifier, which makes per-feed power constraints the most natural formulation from an implementation perspective. However, owing to the additional complexity associated with per-feed power constraints, most existing precoding techniques are formulated under a total transmit-power constraint. In this dissertation, the emphasis is placed on sum-rate maximization subject to a total transmit power constraint, in line with the terrestrial MU-MIMO framework developed in the previous chapters. A further classification axis is the system architecture. In large multibeam systems, precoding may be distributed across multiple GWs, and part of the processing may even be transferred on board the satellite in order to relax feeder-link limitations. Here, the simplest architecture is considered: a single GW performs all precoding on the ground and serves the multibeam payload through a common feeder link.

Finally, and most importantly for the present work, precoding techniques differ in the type of CSI assumed at the transmitter. Three regimes are commonly distinguished: perfect instantaneous CSI, statistical CSI, and imperfect or outdated CSI. The vast majority of the satellite precoding literature assumes perfect instantaneous CSI at the GW. This assumption becomes unrealistic in the scenario considered here, where the combination of large propagation delay and a time-varying channel due to a UT-induced Doppler renders the CSI available at the GW systematically outdated. Precoding under outdated CSI is therefore the regime adopted in this work.

Related Work and Positioning

To the best of our knowledge, only two works have addressed the design of linear precoding for mobile multi-beam satellite systems with outdated CSI under assumptions comparable to ours.

The first is the work of Vázquez et al. [7], which considers an L-band mobile interactive multi-beam system over a GEO satellite. The authors model the feedback loop as a single delay τ on the order of 700 milliseconds and adopt a simple closed-form minimum mean squared error (MMSE) precoder. Their key analytical observation is that, for a fixed precoder, the random phase fluctuations of the fading process affect the desired and the interfering terms symmetrically and therefore cancel in the resulting signal-to-interference-plus-noise ratio (SINR), only the amplitude fluctuations of the fading process remain visible. As a consequence, the ergodic sum rate is essentially unchanged by outdated CSI, and the dominant practical impairment becomes the mismatch between the SINR predicted by the outdated CSI and the SINR effectively experienced by the UTs, which is handled by fixed back-off margins in the link-adaptation stage.

The second is the work of Joroughi et al. [8], which considers a maritime multi-beam scenario at L-band. The channel is decomposed into a time-invariant beam-gain matrix and a time-varying diagonal attenuation matrix, and the authors propose two CSI feedback strategies. The first is a memoryless strategy that feeds the GW with the latest delayed snapshot of the attenuation.

The corresponding precoder is a conventional zero-forcing (ZF) or MMSE precoder computed from this estimate. The second is a memory-based strategy that averages the attenuation over several past snapshots, and it is paired with a novel two-stage precoder: a first stage suppresses inter-beam interference, while a second diagonal stage compensates for the residual time-varying attenuation.

While the works above tackle the same regime of outdated CSI in mobile multi-beam satellite systems, our approach differs in ...

Chapter 2

Precoding under Outdated CSI

In Chapter ??, linear precoding was developed under the implicit assumption that the CSI used to design the precoder coincides with the CSI experienced by the UTs at the moment of transmission. As discussed in Chapter 1, this assumption breaks down in the mobile satellite scenario considered in this work, owing to the combination of a large propagation delay and a UT-induced Doppler spread. This chapter is devoted to characterizing the impact of this mismatch on precoder performance and to developing a strategy that compensates for it.

Section 2.1 formulates the problem precisely and introduces the CSI feedback delay as the key operating parameter. From there, the chapter proceeds in three steps. First, Section 2.2 quantifies the impact of outdated CSI on the achievable rate of the linear precoders developed in Chapter ?. Section 2.3 then proposes a compensation method that exploits the temporal correlation of the channel in order to predict the instantaneous channel at the time of transmission. Finally, Section 2.4 re-evaluates both the impact of outdated CSI and the proposed compensation strategy under a more realistic channel model.

To end this work, the results are applied to a set of representative example scenarios, after which the main findings are summarized in a final conclusion.

2.1 Problem Formulation

We consider the forward link of the multi-beam HTS system as described in Chapter 1. The GW serves K UTs simultaneously through the satellite, equipped with N_T feeds, by applying a linear precoder to the data symbols. Each UT is equipped with N_R receive antennas. The simulations performed in this chapter focus on a system in which the BS has 8 transmit antennas and serves 4 UTs with 2 receive antennas each. The channel of UT k at time t is denoted by $\mathbf{H}_k(t) \in \mathbb{C}^{N_R \times N_T}$ and follows the Rician model from (1.1). The received signal at UT k reads

$$\mathbf{y}_k(t) = \mathbf{H}_k(t) \mathbf{F}(t) \mathbf{a}(t) + \mathbf{n}_k(t), \quad (2.1)$$

where $\mathbf{a}(t)$ collects the data symbols of each stream intended for UT k into a column vector, and $\mathbf{n}_k(t) \sim \mathcal{CN}(\mathbf{0}, N_0 \mathbf{I}_{N_R})$ is additive noise. The precoder satisfies the total transmit-power constraint $\text{tr}(\mathbf{F}(t)\mathbf{F}^H(t)) \leq P_T$ at any moment in time.

In practice, the channel must first be estimated at the UTs and fed back to the GW, before it can be used to design the precoder. Denoting the aggregate elapsed time by the CSI feedback delay τ_{CSI} , the precoder applied at time t is necessarily based on the outdated channel state $\mathbf{H}(t - \tau_{\text{CSI}})$. How severely this mismatch degrades performance depends on how much the channel has evolved during τ_{CSI} .

The temporal variability of the channel is characterised through the normalised autocorrelation function of its NLoS component, denoted by $R_h(\tau)$. It is shown in Appendix ?? that, under Jakes' model, the autocorrelation between two channel realizations separated by a lag τ equals the zeroth-order Bessel function of the first kind, evaluated in $2\pi f_D \tau$. The coherence time of the NLoS component is defined as the lag at which the autocorrelation first falls to one half of its peak value:

$$T_c \triangleq \max\{\tau > 0 : R_h(\tau) \geq \frac{1}{2}\}. \quad (2.2)$$

Rather than studying the impact of τ_{CSI} in absolute terms, the CSI feedback delay is expressed as a fraction of the coherence time throughout the analysis. Two operating points sharing the same τ_{CSI}/T_c exhibit, by construction, the same correlation between the outdated and the instantaneous channel, irrespective of the absolute values of τ_{CSI} and T_c . Reporting results as a function of τ_{CSI}/T_c therefore yields conclusions that apply to any combination of carrier frequency, UT velocity and orbital altitude consistent with the underlying Rician model.

2.2 Impact of Outdated CSI

In order to quantify the impact of outdated CSI on the system performance, the achievable rate obtained by linear precoders developed in Chapter ?? is simulated as a function of the signal-to-noise ratio (SNR). Each precoder is designed based on the outdated channel state $\mathbf{H}(t - \tau_{\text{CSI}})$ and applied to the instantaneous channel $\mathbf{H}(t)$. The resulting sum rate is averaged over more than 100 channel realizations, and over multiple coherence periods of the fading process for each realization. Figures 2.1 and 2.2 report the results for a CSI feedback delay τ_{CSI} that varies from $0 T_c$ (instantaneous CSI, used as a reference) up to $5/4 T_c$, slightly beyond the coherence time.

All three precoders are very sensitive to outdated CSI. Even a delay of only a quarter of the coherence time causes a substantial loss on the achievable rate at moderate and high SNR values. The loss with respect to instantaneous CSI grows monotonically with τ_{CSI}/T_c , confirming the intuition that the more outdated the CSI, the worse the performance. Once the CSI feedback delay surpasses the coherence time of the NLoS component, the system becomes essentially unusable. This aligns with the definition of T_c . Beyond the point $\tau_{\text{CSI}} \geq T_c$, the channel available at the GW and the channel actually experienced by the UTs are only weakly correlated such that the precoder is in effect designed for a channel that bears little resemblance to the true one. As a

consequence, the precoder fails to exploit the spatial dimension of the channel and the sum-rate collapses.

A closer look at Figure 2.1 shows that both ZF and block diagonalization (BD) precoding saturate at high SNR once the CSI is even slightly outdated. Both ZF and BD precoders completely eliminate the interference between the users, no matter the ratio of the transmit power to the noise power. With outdated CSI, the precoder nulls interference along the wrong directions, leaving a residual inter-user interference that is proportional to the transmit power. As the SNR increases, the system transitions from a noise-limited regime to an interference-limited one and the desired signal and the residual interference grow at the same rate. Therefore, the SINR saturates and the achievable rate converges to a finite ceiling, that is determined solely by the magnitude of the channel evolution over the feedback delay.

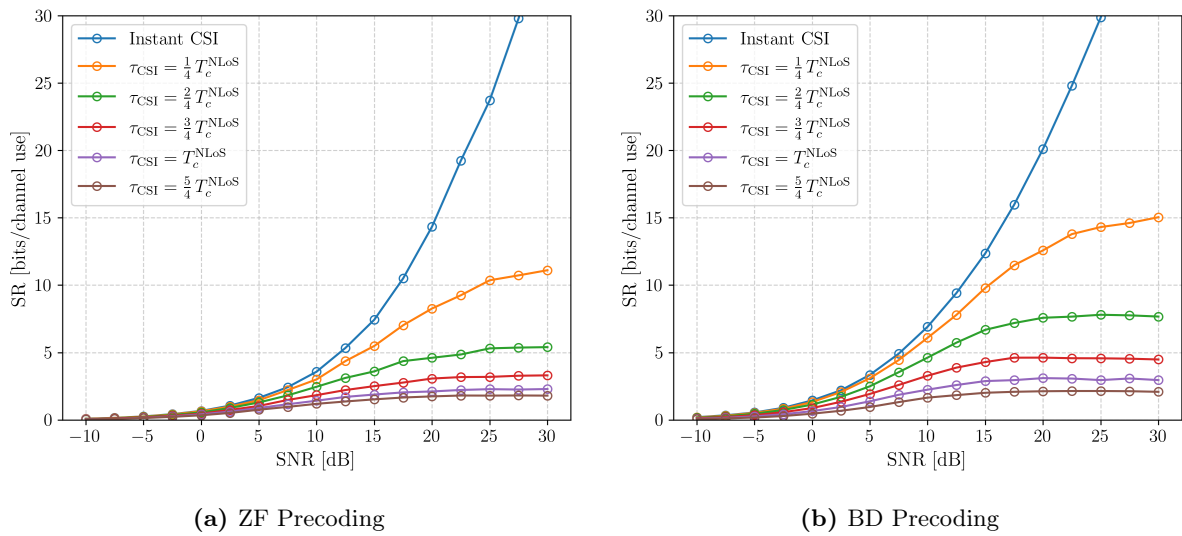


Figure 2.1: The achievable sum-rate of a $8 \times \{2, 2, 2, 2\}$ MU-MIMO system as a function of the SNR, for ZF and BD precoding. The different curves correspond to different values of the CSI feedback delay τ_{CSI} , expressed as a fraction of the coherence time T_c .

The weighted minimum mean squared error (WMMSE) precoder exhibits a qualitatively different behaviour. The achievable rate reaches a maximum and decreases slightly beyond it when the precoder design relies on outdated CSI, as is clearly visible in Figure 2.2. Unlike ZF and BD, WMMSE precoding adapts its solution to the operating SNR. At low SNR it trades interference suppression against desired-signal power, while at high SNR the regularization vanishes and it converges to the interference-free BD solution. The algorithm then increasingly concentrates power in directions that it expects to be beneficial, but which, due to the channel mismatch, actively generate inter-user interference in the actual channel. The ZF-type precoders do not exhibit this behaviour because they steer in the wrong direction just as aggressive for any SNR value.

A final observation concerns the reference curve obtained with instantaneous CSI. Even in this idealised case, the achievable rate lies noticeably below the values obtained in Chapter ?? under a purely Rayleigh fading channel. This accounts for all three precoding techniques, and is a direct consequence of the LoS contribution in a Rician channel. Indeed, the LoS component is

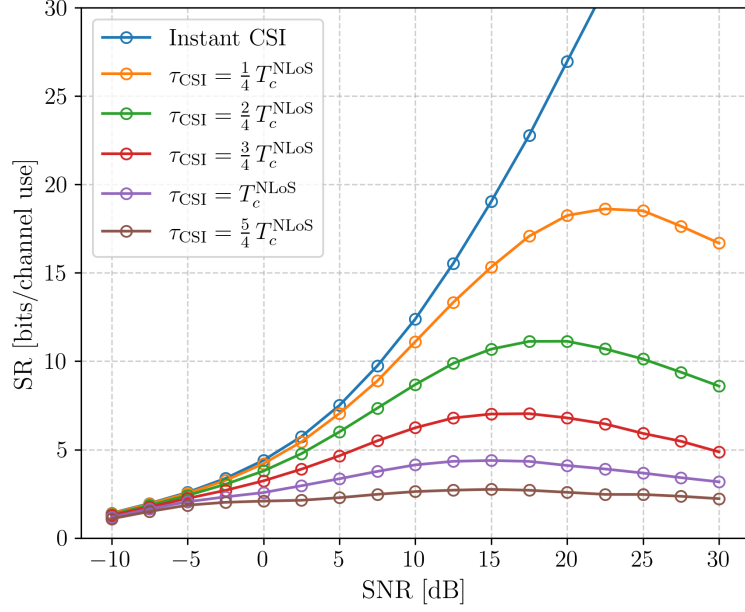


Figure 2.2: Similar figure as Fig. 2.1, but for WMMSE precoding.

rank-one, such that the largest fraction of the channel energy is concentrated along a single spatial direction. A purely Rayleigh fading channel, by contrast, provides rich spatial diversity and a well-conditioned channel matrix, granting the precoder significantly more spatial multiplexing gain. Therefore, the LoS component imposes an inherent limit on the spatial degrees of freedom available, regardless of the quality of the CSI.

Taken together, these results show that even a small feedback delay poses a threat to the achievable rate at moderate and high SNR, and that no choice of precoder can restore the linear growth observed with instantaneous CSI by itself. The next section therefore turns to the design of a compensation strategy that targets the source of the problem, which is the temporal mismatch between $\mathbf{H}(t - \tau_{\text{CSI}})$ and $\mathbf{H}(t)$, rather than its consequences.

2.3 Compensation for Outdated CSI

As established in the previous section, the degradation observed in Section 2.2 stems from the mismatch between the channel estimate available at the GW and the channel experienced at the time of transmission. However, the channel is a temporally correlated random process, and the history of past channel estimates therefore carries useful information about the current channel state. In what follows, we propose a simple compensation strategy that exploits the temporal correlation to predict, at each UT, the channel that will be experienced after the CSI feedback delay, and feed this predicted channel back to the GW instead of the most recent channel estimate. The performance gain introduced by this strategy will again be evaluated by means of a simulation.

2.3.1 Compensation Strategy

Among the many possible prediction methods, a linear autoregressive (AR) model is adopted here as a simple and tractable baseline. It leads to a closed-form predictor, requires only knowledge of the autocorrelation function, and entails a very limited computational complexity. The prediction is performed locally at each UT, while the GW continues to compute the precoder exactly as in Chapter ???. From the perspective of the GW, the precoding procedure is therefore unchanged. Only the channel information on which it relies, is replaced by a prediction of the future channel state. As a result, the additional complexity introduced by the compensation strategy is borne entirely by the UTs.

Under the simplified channel model of Chapter 1, the temporal variation is entirely contained in the NLoS component. Moreover, spatial correlation within the NLoS component is neglected, so that the entries of $\mathbf{H}_k^{\text{NLoS}}(t)$ evolve independently and share the same temporal correlation function. Therefore, we choose to carry out the prediction element-wise, using the same scalar predictor for each channel coefficient. Let $\hat{h}_{k,(i,j)}[n]$ denote the estimate of the (i,j) -th entry of the channel matrix of UT k at the n -th pilot instant. A p -th order AR predictor of the channel Δ pilot intervals ahead is then written as

$$\hat{h}_{k,(i,j)}[n + \Delta] = \sum_{m=0}^{p-1} \alpha_m \hat{h}_{k,(i,j)}[n - m]. \quad (2.3)$$

The prediction horizon Δ is chosen such that the predicted sample corresponds to the actual transmission time, that is, $\Delta T_p = \tau_{\text{CSI}}$, where T_p denotes the pilot period.

The predictor coefficients $\{\alpha_m\}_{m=1}^p$ are obtained from the Yule–Walker equations associated with the autocorrelation function $R_h(\tau)$ of the channel process:

$$\begin{bmatrix} R_h(0) & R_h(T_p) & \cdots & R_h((p-1)T_p) \\ R_h(T_p) & R_h(0) & \cdots & R_h((p-2)T_p) \\ \vdots & \vdots & \ddots & \vdots \\ R_h((p-1)T_p) & R_h((p-2)T_p) & \cdots & R_h(0) \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_p \end{bmatrix} = \begin{bmatrix} R_h(\Delta T_p) \\ R_h((\Delta+1)T_p) \\ \vdots \\ R_h((\Delta+p-1)T_p) \end{bmatrix}. \quad (2.4)$$

Since the autocorrelation function is fully specified by the assumed channel model, these coefficients can be computed offline once the predictor order p , the pilot period T_p , and the delay τ_{CSI} have been fixed.

Three assumptions underlie the proposed compensation strategy. First, channel-estimation errors are neglected, so that the channel estimates available at the UTs are assumed to be perfect. Second, each UT is assumed to know the second-order statistics of its channel. In the present model, this amounts to knowing the autocorrelation function $R_h(\tau)$, which under Jakes' model is fully determined by the maximum Doppler frequency f_D . This is a reasonable assumption, since f_D can be estimated locally from the carrier frequency and the UT velocity. Third, the pilot period T_p is treated as a free design parameter and is not constrained by a pilot-overhead budget. This idealization makes it possible to characterize the intrinsic potential of the compensation strategy independently of the resource limitations of a specific system implementation.

2.3.2 Parameter Optimization

The predictor of (2.3) is controlled by two free parameters. Firstly, the pilot period T_p , or equivalently the pilot rate $1/T_p$, which determines the temporal resolution at which the channel is observed. Secondly, the time span or context window length $L \triangleq p T_p$ covered by the p past samples, which controls how far back in time the predictor reaches. For a fixed window length, a higher pilot rate raises the model order p by providing finer temporal resolution, while for a fixed pilot rate, a longer window increases the model order by incorporating older observations. The order p of the AR model is fixed once T_p and L are chosen through the relation $p = L/T_p$, and thus not considered as a third design parameter. Both model parameters are expressed relative to the coherence time of the channel, in line with the normalization of the CSI feedback delay adopted throughout this chapter.

Rather than performing a joint grid search over T_p and L , the two parameters are tuned sequentially. The pilot rate is first optimised with the context window length held fixed at a conservative value, after which the context window length is tuned with the pilot rate frozen at the value found in the first step. Both optimisations are evaluated for the $8 \times \{2, 2, 2, 2\}$ system using WMMSE precoding at a single operating point with an SNR equal to 20 dB, for different values of CSI feedback delay.

For the first sweep, the context window length fixed at $2T_c$. This is a generous choice, made to ensure that no potentially useful information would be excluded. Figure 2.3 shows the achievable sum-rate as a function of the pilot rate for four values of the CSI feedback delay. The achievable sum-rate increases steadily as the pilot rate grows, reflecting the improvement in temporal resolution available to the predictor. When too few samples are available within the temporal correlation window to capture the channel dynamics accurately, the achievable sum-rate drops rapidly, especially for the larger feedback delays. At 6 messages per T_c , all four curves reach their respective maximum and converge to nearly the same value, indicating that the compensation is

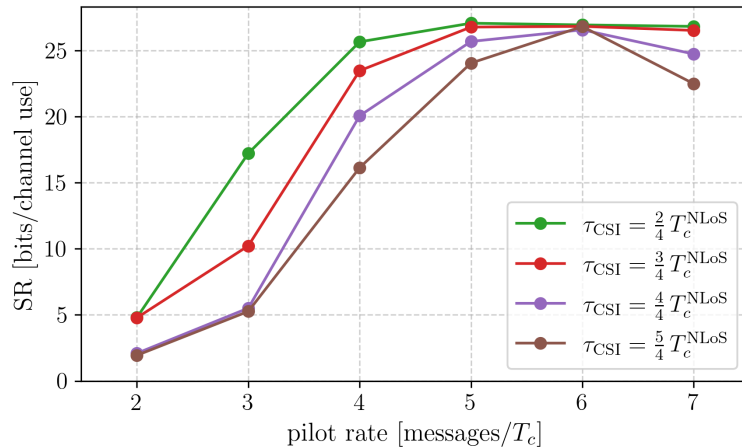


Figure 2.3: The achievable sum-rate plotted as a function of the pilot rate. The $8 \times \{2, 2, 2, 2\}$ system uses a WMMSE precoder and AR-based channel prediction model at the UT, and operates at a SNR equal to 20 dB. The context window length is fixed to $2T_c$. Each curve corresponds to a different CSI feedback delay.

equally effective regardless of the feedback delay. Increasing the pilot rate further than 6 messages per coherence time does not improve performance and even causes a slight degradation for the larger delays, most likely because the resulting high model order renders the Yule-Walker system ill-conditioned. The autocorrelation matrix then becomes increasingly close to singular. A pilot rate of 6 messages per coherence time is therefore selected.

With the pilot rate fixed at 6 messages per T_c , the context window length is varied. Figure 2.4 shows that a too short window prevents the predictor from accessing enough of the temporal correlation of the channel and, as expected, severely degrades the achievable sum-rate. Also, the curves saturate beyond a certain prediction horizon. This aligns with the intuition that, after a certain point, the channel history is no longer significantly correlated with the value being predicted, such that the additional information contained in older samples is negligible. This saturation point depends strongly on the CSI feedback delay: for shorter feedback delays, a shorter window length suffices. Note that a longer context window length results in a larger memory footprint at the UT. A context window length of one coherence period is therefore adopted for all subsequent simulations.

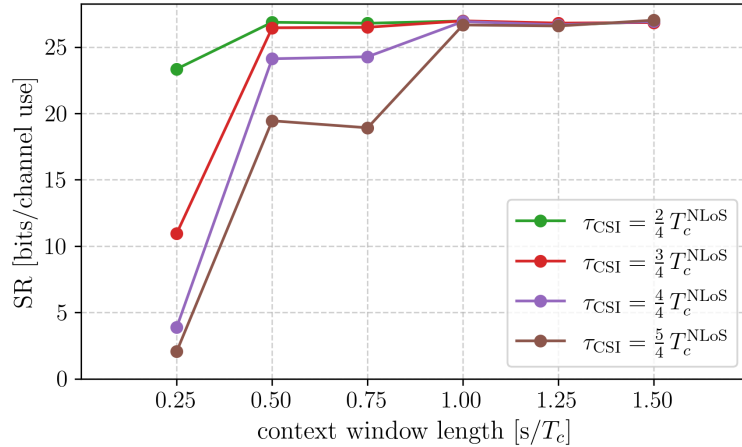
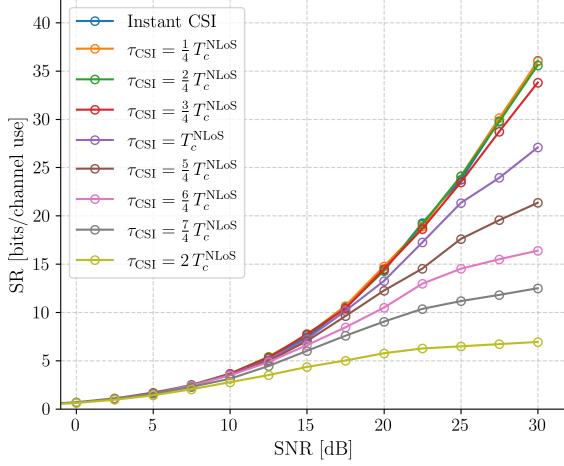


Figure 2.4: The achievable sum-rate plotted as a function of the context window length. The pilot rate is set to 6 messages per coherence period. Other than that, the system operates under the exact same configurations in Fig. 2.3.

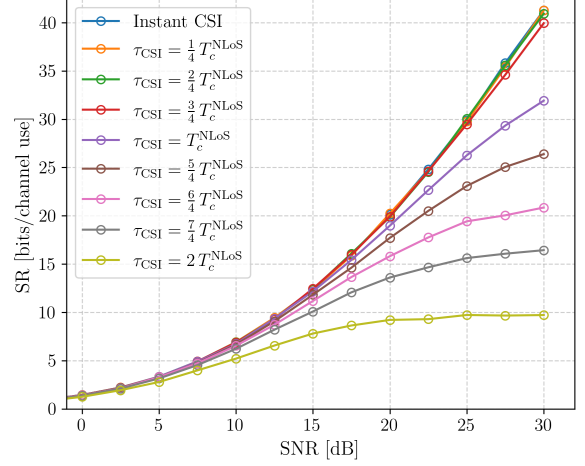
2.3.3 Performance Evaluation

This subsection quantifies the gain achieved by the proposed AR-based compensation strategy. The predictor settings are those selected in Section 2.3.2: a pilot rate of 6 messages per coherence period and a context window length equal to one coherence period. All other system configurations are kept identical to Section 2.2, so that the effect of compensation can be isolated directly. Figure 2.5 shows the achievable sum-rate for ZF and BD precoding, and Figure 2.6 provides an explicit three-way comparison for WMMSE precoding. Each panel shows the achievable sum-rate under perfect CSI, outdated CSI, and predicted CSI for a single feedback delay.

Figures 2.5 and 2.6 show a consistent trend across all three precoding techniques. For feedback delays up to approximately one coherence time, the compensation mechanism restores the



(a) ZF Precoding



(b) BD Precoding

Figure 2.5: The achievable sum-rate as a function of the SNR for ZF and BD precoding with AR-based channel prediction at the UTs. The predictor uses the parameter pair selected in Section 2.3.2. The different curves correspond to different CSI feedback delays, from no delay up to a delay equal to 2 coherence periods of the NLoS component.

performance obtained with instantaneous CSI almost completely. This confirms that the dominant part of the mismatch can be mitigated as long as the predictor operates within a time interval where the NLoS component remains sufficiently correlated. When the feedback delay exceeds the coherence time, the gain gradually decreases. This behaviour is expected: for $\tau_{\text{CSI}} > T_c$, the channel samples available to the predictor become weakly correlated with the target value, so the predicted channel cannot accurately track the channel realization experienced at the time of transmission. The compensation strategy remains beneficial in this regime, but the residual mismatch grows with delay and the achievable sum-rate drops accordingly.

An remarkable difference between Figure 2.1 and Figures 2.5 is the disappearance of the rate ceiling. The compensation strategy restores the growth of the achievable sum-rate with increasing SNR for almost all considered CSI feedback delays. Only when τ_{CSI} approaches $2T_c$, the residual prediction error is large enough to reintroduce an interference floor, and the familiar saturation behaviour of §2.2 reappears. A similar observation can be made for WMMSE precoding. Only for feedback delays well above T_c , the sum-rate exhibits the peak-and-decrease behaviour identified in Section 2.2.

Altogether, these results show that channel prediction at the UTs is an effective and low-complexity mitigation mechanism for CSI aging under the Rician fading channel model. The largest performance gains are obtained when the feedback delay is on the order of, or smaller than, the coherence time of the NLoS component.

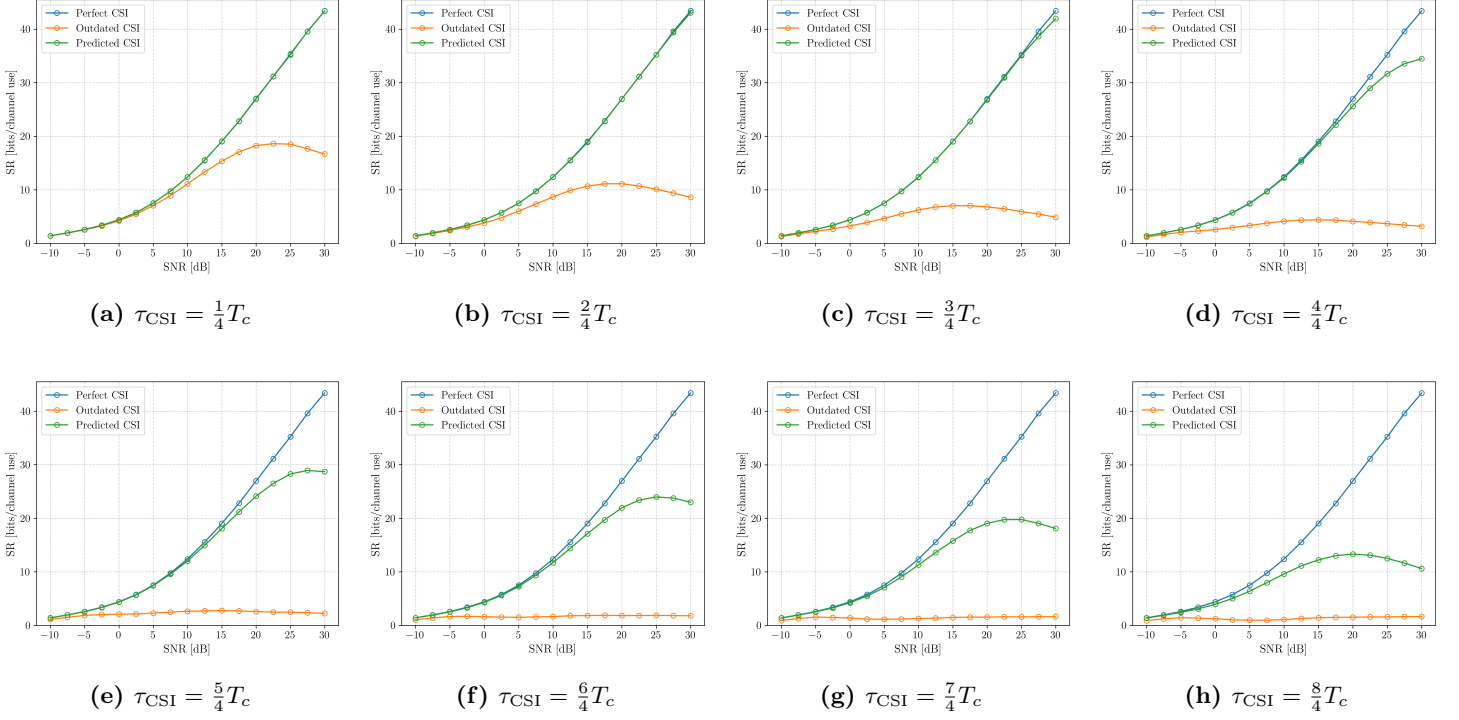


Figure 2.6: A comparison of the WMMSE precoding performance under perfect CSI, outdated CSI, and compensated CSI obtained through AR-based prediction at the UTs. The compensation gain is strongest for $\tau_{\text{CSI}} \leq T_c$ and decreases progressively as the feedback delay exceeds the channel coherence time.

2.4 Evaluation on a More Realistic Channel

2.4.1 Channel Model

The Rician model used in Sections 2.2 and 2.3 provides a useful baseline, but it neglects two physically relevant effects in practical mobile satellite links. First, the antenna array geometry at both the satellite and the UTs and the propagation directions introduce spatial correlation between channel coefficients. Second, the motion of a mobile UT induces a deterministic Doppler shift on the LoS component that cannot always be pre-compensated. The ray-tracing based channel model introduced in this section extends the Rician model and incorporates both effects. It is based on the channel model from [9, 10, 11].

Both the satellite and each UT are equipped with a uniform planar array (UPA), as illustrated in Figure 2.7. The satellite array has $N_T = N_T^x \times N_T^y$ elements, with inter-element spacings d_T^x and d_T^y along the x- and y-axis, respectively. Fully analog, the UT array has $N_R = N_R^x \times N_R^y$ elements, with spacings d_R^x and d_R^y . For a wavefront arriving from elevation angle θ and azimuth angle ϕ , the array steering vector of a UPA with $N = N_x \times N_y$ elements reads

$$\mathbf{a}(\theta, \phi)_{(n_x N_y + n_y)} = \frac{1}{\sqrt{N}} \exp\left(j \frac{2\pi}{\lambda} (n_x d_x \sin \theta \cos \phi + n_y d_y \sin \theta \sin \phi)\right), \quad (2.5)$$

for $n_x = 0, \dots, N_x - 1$ and $n_y = 0, \dots, N_y - 1$, where λ is the carrier wavelength.

Because the scattering on ground takes place only within a few kilometers around each UT, the angle of departure (AoD) for different propagation paths to the same UT are nearly identical

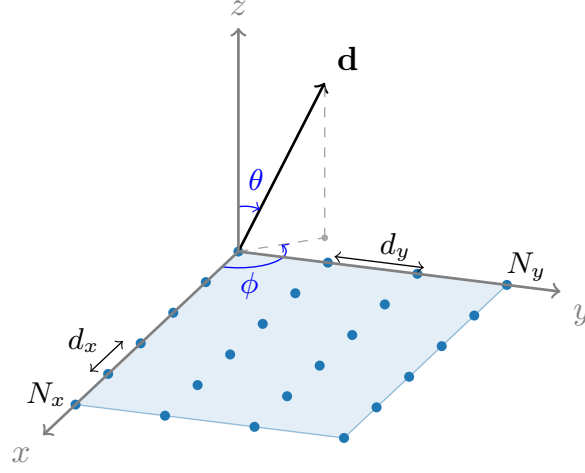


Figure 2.7: The uniform planar array (UPA) antenna configuration considered at both the satellite and the UTs. The vector $\mathbf{d}$ points in the satellite-to-user direction.

due to the exceptional propagation distance in satellite communications [10]. Therefore, a single satellite-side steering vector $\mathbf{a}_k^{\text{sat}} = \mathbf{a}(\theta_k^{\text{sat}}, \phi_k^{\text{sat}}) \in \mathbb{C}^{N_T}$ applies to every component of the channel, where $(\theta_k^{\text{sat}}, \phi_k^{\text{sat}})$ is the elevation-azimuth angle pair to UT k from a satellite point-of-view. By contrast, from a UT perspective scattering is local, so each ray within a cluster arrives from a distinct direction described by an individual steering vector $\mathbf{a}_{k,c,\ell}^{\text{ut}} = \mathbf{a}(\theta_{k,c,\ell}^{\text{ut}}, \phi_{k,c,\ell}^{\text{ut}}) \in \mathbb{C}^{N_R}$.

The channel matrix for UT k is modelled as

$$\begin{aligned} \mathbf{H}_k(t) &= \sqrt{\frac{\kappa}{\kappa+1}} \mathbf{H}_k^{\text{LoS}}(t) + \sqrt{\frac{1}{\kappa+1}} \mathbf{H}_k^{\text{NLoS}}(t) \\ \text{with } \mathbf{H}_k^{\text{LoS}}(t) &= \alpha_k^{\text{LoS}}(t) \mathbf{a}_k^{\text{ut}} (\mathbf{a}_k^{\text{sat}})^{\text{H}} \\ \text{and } \mathbf{H}_k^{\text{NLoS}}(t) &= \sum_{c=1}^C \sqrt{w_c} \cdot \frac{1}{\sqrt{L_c}} \sum_{\ell=1}^{L_c} \alpha_{k,c,\ell}^{\text{NLoS}}(t) \mathbf{a}_{k,c,\ell}^{\text{ut}} (\mathbf{a}_k^{\text{sat}})^{\text{H}}, \end{aligned} \quad (2.6)$$

where κ is the Rician factor, c indexes the NLoS scattering clusters with relative powers w_c , and L_c is the number of rays in cluster c .

The LoS gain $\alpha_k^{\text{LoS}}(t)$ incorporates the Doppler shift induced by the UT's motion:

$$\alpha_k^{\text{LoS}}(t) = \exp(j(\psi_k + 2\pi\nu_k^{\text{LoS}} t)), \quad (2.7)$$

where $\psi_k \sim \mathcal{U}(0, 2\pi)$ is a random initial phase and $\nu_k^{\text{LoS}} = f_D \cos(\pi/2 - \theta_k^{\text{sat}})$ is the Doppler frequency of the LoS path. Here, $f_D = v_{\text{UT}} f_c / c$ is the maximum Doppler frequency, determined by the carrier frequency f_c , the UT velocity v_{UT} , and the speed of light c . The factor $\sin \theta_k^{\text{sat}}$ accounts for the projection of the UT's horizontal velocity onto the satellite-to-user direction. Unlike the simplified model of (1.1), the LoS component is no longer time-invariant. It rotates in phase at a rate that depends on the UT's velocity and elevation angle.

Each cluster consists of L_c rays whose temporal evolution follows the same Jakes isotropic-scattering model as in the simplified channel of Section 2.1.

$$\alpha_k^{\text{NLoS}}(t) \sim \mathcal{CN}(0, 1) \quad \text{with} \quad R_{\alpha_k^{\text{NLoS}}}(\Delta t) = J_0(2\pi f_D \Delta t). \quad (2.8)$$

The angle of arrival (AoA) at UT k of a ray ℓ within cluster c is

$$\theta_{k,c,\ell}^{\text{ut}} = \theta_k^{\text{ut}} + \Delta\theta_{k,c} + \sigma_\theta u_{k,c,\ell}, \quad (2.9)$$

$$\phi_{k,c,\ell}^{\text{ut}} = \phi_k^{\text{ut}} + \Delta\phi_{k,c} + \sigma_\phi v_{k,c,\ell}, \quad (2.10)$$

where $(\theta_k^{\text{ut}}, \phi_k^{\text{ut}})$ is the LoS direction as seen at the UT, $(\Delta\theta_{k,c}, \Delta\phi_{k,c})$ is the mean angular offset from the LoS direction for cluster c , σ_θ and σ_ϕ are the intra-cluster angular spreads, and $u_{k,c,\ell}$, $v_{k,c,\ell}$ are independent Laplace-distributed random variables with zero mean and unit variance. We restrict ourselves to 2 clusters. The first cluster is centred on the LoS direction ($\Delta\theta_{k,1} = \Delta\phi_{k,1} = 0$), while the second cluster is offset by a user-specific, randomly drawn displacement $(\Delta\theta_{k,2}, \Delta\phi_{k,2})$. The elevation angle of the LoS path is the same at both the satellite and the UT, i.e. $\theta_k^{\text{ut}} = \theta_k^{\text{sat}}$. The orientation of the UT's array, i.e. ϕ_k^{ut} , relative to the satellite's array is determined by a rotation ρ around the vertical axis, which is uniformly distributed in $[0, 2\pi)$ and independent of the satellite-side azimuth.

Since every term in (2.6) shares the same satellite-side steering vector $\mathbf{a}_k^{\text{sat}}$, the channel matrix of a single UT is of rank one at any moment in time. Consequently, regardless of the number of receive antennas, the channel of each UT supports at most one spatial stream. This significantly alters the behavior of the precoding schemes considered in this work, since the receive-side spatial multiplexing gain is no longer available. Because the purpose of this section is not to study this rank limitation itself, the subsequent evaluation is restricted to system configurations in which each UT is equipped with a single receive antenna.

A second distinction from the simplified model concerns the spatial structure of the channel. Under the ray-based model of (2.6), the entries of the channel matrix for UT k are no longer spatially independent. In principle, a predictor that exploits this spatial correlation jointly with the temporal one could yield more accurate predictions than the element-wise approach of Section 2.3.1. However, developing such a predictor would significantly increase the complexity at each UT. To maintain a fair and direct comparison with the results of Section 2.3, the same element-wise AR predictor is applied here without modification, with the same parameter values established in Section 2.3.2.

2.4.2 Performance Evaluation

In this section, we perform a first evaluation of the bit error rate (BER) of an $8 \times \{1, 1, 1, 1\}$ system using a ZF precoder. Each UT is equipped with a single receive antenna and is served with one data stream carrying 16-quadrature amplitude modulation (QAM) symbols. In analogy with Section 2.2, the CSI feedback delay is expressed as a fraction of the coherence time T_c of the NLoS component, as defined in (2.2), so that the results remain independent of any particular choice of UT velocity or carrier frequency.

The left panel of Figure 2.8 shows the BER under outdated CSI as a function of the SNR for several values of the feedback delay. The qualitative behavior closely mirrors the observations made in Section 2.2 for the simplified Rician model. As the feedback delay grows, the BER at any given SNR degrades monotonically. For delays approaching or exceeding T_c^{NLoS} , the residual inter-user interference caused by the channel mismatch begins to dominate the noise at high SNR, and the BER curves flatten. The system has an extremely poor reliability, no matter the available transmit power.

The right panel of Figure 2.8 shows the same system with AR-based channel prediction. The compensation restores a substantial fraction of the performance lost due to the feedback delay, in line with the findings of Section 2.3.3. A notable difference with respect to the simplified model, however, is that the predictor remains effective over a wider range of CSI feedback delays. This extended compensation range can be attributed to the deterministic LoS component. Because the LoS phase rotates at a fixed rate ν_k^{LoS} that is fully determined by the satellite geometry and the UT velocity, the AR model can track this contribution accurately even when the coherence time of the NLoS component has been exceeded.

These first results are promising: the AR-based compensation strategy, developed and tuned under the simplified Rician model, is capable of significantly reducing the BER for feedback delays up to 4 times the NLoS coherence time. At the same time, these simulations only scratch the surface of what a dedicated analysis of this channel model could reveal. A thorough parameter study, an investigation of the spatial and temporal correlation structure on precoding performance, and a comparison across multiple precoding strategies are needed to draw sound conclusions, and are left as natural directions for future work.

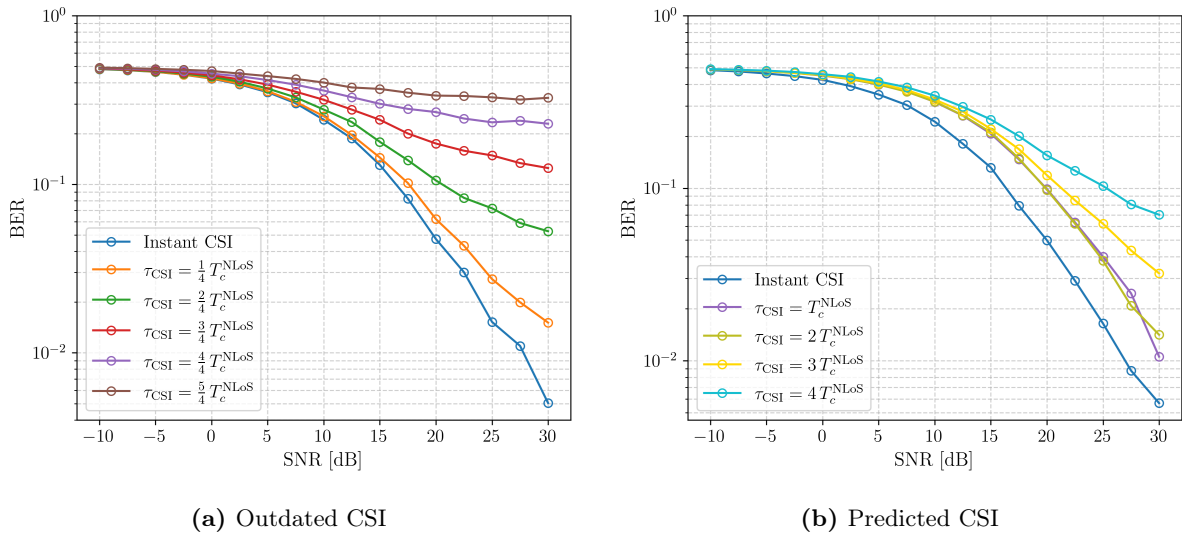


Figure 2.8: title

2.5 Conclusion

$f_c = 1.5$ GHz (mobile satellite services operate mostly in the L-band) $c = 299\,792\,458$ m/s (speed of light)

v | 5 km/h | 15 km/h | 20 km/h | 30 km/h | 40 km/h | fD | 6.95 Hz | 20.85 Hz | 27.80 Hz | 41.70 Hz | 55.59 Hz | Tc NLoS | 34.838 ms | 11.613 ms | 8.709 ms | 5.806 ms | 4.355 ms |

Example 1: LEO satellite. $d = 550$ km $\tau_{\text{propagation}} = 1.835$ ms $\tau_{\text{CSI}} = 7.34$ ms (4 times $\tau_{\text{propagation}}$, because first pilots are sent, estimate/prediction sent back to GW through BS, GW send precoder to satellite) conclusion : under Rician channel model, the achievable can be completely restored for slowly moving UTs up to 15 km/h. Also for higher velocities, the AR-based compensation gives a good/decent result, when just using outdated CSI would give an extreme poor performance.

Example 2: GEO satellite. $d = 35\,786$ km $\tau_{\text{propagation}} = 119.37$ ms $\tau_{\text{CSI}} = 477.48$ ms For a GEO satellite, the propagation delay is of order way larger than for LEO, which makes our compensation strategy useless in this scenario. Even if we assume that all digital processing can happen on-board which is not very realistic atm, such that it is not needed to go to the GW and back, such that τ_{CSI} would equal 2 times the propagation delay instead of 4, the delay is still way too large. In this scenario, we are advised to use other robust precoding techniques.

2.6 Conclusion

This chapter addressed a major limitation of linear precoding in mobile satellite systems: the CSI used at the GW is inevitably outdated at transmission time. After restating the problem in terms of the normalized delay τ_{CSI}/T_c , we quantified the resulting mismatch, proposed a compensation strategy based on channel prediction at the UTs, and evaluated its robustness under a more realistic ray-based satellite channel model.

The analysis of Section 2.2 showed that all considered precoders are highly sensitive to CSI aging. Even moderate delay values cause a substantial loss at medium and high SNR, and once τ_{CSI} approaches the coherence time of the NLoS component, the system becomes strongly interference-limited.

To mitigate this effect, Section 2.3 introduced an element-wise AR predictor at the UTs, with unchanged precoding processing at the GW. After tuning the pilot rate and context window, the proposed strategy was shown to recover most of the performance loss for delays up to about one coherence time, and to remain beneficial beyond that range. Hence, channel prediction offers a practical and low-complexity way to extend the operating region of linear precoding under delayed CSI feedback.

Section 2.4 then replaced the simplified Rician model by a ray-based model including spatial correlation and residual LoS Doppler. The same compensation mechanism still provided clear performance gains. The additional limitation caused by the user channel being rank-deficient was avoided by restricting the evaluation to single-antenna UTs.

Finally, let us map the normalized-delay results to representative mobile satellite services (MSS) operating points to provide a more practical insight. The total CSI feedback delay is modelled as four times the one-way propagation delay between the UT and the satellite. This factor of

four accounts for the downlink pilot transmission to the UTs, the transmission of the estimated or predicted channel over the return link from the UT to the GW through the satellite, and finally the transmission of the precoder matrix over the forward feeder link back to the satellite. MSS typically operate in the L -band [1], with carrier frequencies between 1 and 2 GHz, and corresponding wavelengths ranging from 15 to 30 cm. Table 2.1 summarizes the maximum Doppler frequency and the NLoS coherence time for different UT velocities.

UT velocity	f_D	T_c^{NLoS}
5 km/h	6.95 Hz	34.84 ms
15 km/h	20.85 Hz	11.61 ms
20 km/h	27.80 Hz	8.71 ms
30 km/h	41.70 Hz	5.81 ms
40 km/h	55.59 Hz	4.36 ms

Table 2.1: The maximum Doppler frequency f_D and NLoS coherence time T_c^{NLoS} as a function of UT velocity, for a carrier frequency in the L-band ($f_c = 1.5$ GHz).

For a LEO satellite at an altitude of 550 km, the one-way propagation delay is $\tau_{\text{prop}} = 1.84$ ms, which results in a total CSI feedback delay of $\tau_{\text{CSI}} = 7.34$ ms. Comparing this value against Table 2.1, the condition $\tau_{\text{CSI}} \leq T_c$ is satisfied for all UT moving slower than approximately 20 km/h. In this regime, the AR-based compensation operates at its most effective and the achievable rate is restored to near-perfect-CSI performance. For higher velocities, such as 30 km/h or 40 km/h, the feedback delay exceeds the coherence time and perfect restoration is no longer possible. However, the compensation still provides a substantial gain: whereas uncompensated outdated CSI would render the system essentially unusable at these speeds, the predicted CSI preserves a meaningful link quality. The AR-based strategy is therefore well-suited to LEO satellite scenarios across a wide range of UT velocities, under the assumption of a Rician fading channel as in 1.1.

For a GEO satellite at an altitude of 35 786 km, the one-way propagation delay is $\tau_{\text{prop}} = 119.4$ ms, yielding $\tau_{\text{CSI}} = 477.5$ ms. Even for the slowest UT velocity considered (5 km/h, $T_c = 34.84$ ms), this corresponds to $\tau_{\text{CSI}}/T_c \approx 13.7$, far beyond the range where temporal prediction is effective. The AR-based compensation strategy is therefore not applicable to GEO links. In this regime, the mismatch between the available and the actual channel is too large to be bridged the predictor. More robust precoding formulations that explicitly account for severe CSI uncertainty are required.

Part III

Appendices

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